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February 16, 1978
SUPERSEDING
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April 12, 1965

FEDERAL SPECIFICATION

PAINT, BLACKBOARD COATING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a low sheen paint suitable as a chalk board coating used to restore writing surfaces.

1.2 Classification.

1.2.1 Types. Blackboard coating paint covered by this specification shall be furnished in the following types.

Type I - Paint, blackboard coating, black.
Type II - Paint, blackboard coating, green.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

49 CFR 171-189 - Department of Transportation (DOT) Hazardous Materials Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders should cite the latest edition and supplements thereto.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 523 - Test for Specular Gloss.
D 562 - Consistency of Paints Using the Stormer Viscosimeter.
D 822 - Light- and Water-Exposure Apparatus for Testing Paint and Related Products.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems.
D 1296 - Odor of Volatile Solvents and Diluents.
D 1729 - Visual Evaluation of Color Differences of Opaque Materials.
D 2486 - Scrub Resistance of Interior Latex Flat Wall Paints.
D 2805 - Test for Hiding Power of Paints.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. *REQUIREMENTS*

3.1 Composition.

3.1.1 Pigment. Type I. Any suitable black or combination of black and iron blues together with extending pigments may be used in type I, paint, blackboard coating.

3.1.2 Pigment. Type II. Any combination of green pigments plus extender and tinting pigments may be used in type II, paint blackboard coating.

3.1.3 Vehicle. The vehicle for either type shall consist of a modified alkyd resin combined with thinners or an oleoresinous, phenolic-modified vehicle with a suitable thinner.

3.2 Quantitative requirements. The paints shall meet the quantitative requirements specified in table I for the respective types.

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TABLE I.

Characteristics	Requirements			
	Type I black		Type II green	
	Min.	Max.	Min.	Max.
Pigment (percent by weight of paint)	28	-	42	-
Nonvolatile vehicle (percent by weight of vehicle)	27	-	34	-
Consistency, Krebs-Stormer shearing rate, 200 r.p.m.				
Grams	140	220	125	175
Equivalent K. U.	70	85	65	75
Contrast ratio	0.99	-	0.99	-
Specular gloss, 85°	-	15	-	5
Fineness of grind	2	4	2	4
Wt. per gallon, lbs.	8.5	10.5	10	12
Coarse particles and skins (residue retained on a No. 325 sieve) % of total paint	-	1	-	1
Drying time (in hours):				
Dust free	-	1/2	-	1/2
Dry through	-	16	-	16
Water, percent by weight	-	0.5	-	0.5
Lead, percent by weight of nonvolatile	-	0.06	-	0.06

3.3 Qualitative requirements.

3.3.1 Condition in container. The paint shall not settle, cake, or thicken to such a degree that it cannot be easily redispersed with a paddle when tested as specified in table II.

TABLE II. Test procedures

Tests	Fed. Test Method Std. No. 141	ASTM Method	Requirement paragraph	Paragraph reference
Pigment, percent	4021	-	Table I	-
Nonvolatile	4051	-	Table I	-
Consistency	-	D 562	Table I	-
Abrasion and moisture resistance	-	D 2486	3.3.9	4.4.7
Drying time	4061	-	Table I	-
Brushing, flowing and leveling	2141	-	3.3.5	4.4.3
Fineness of grind	-	D 1210	Table I	-
Color	-	D 1729	3.3.6	4.4.4
Condition in container	3011	-	3.3.1	-
Flexibility	6221	-	3.3.2	4.4.2
Odor	-	D 1296	3.3.4	-
Skinning	3021	-	3.3.3	-
Fastness of light	-	D 822-	3.3.7	4.4.5
Specular gloss	-	D 523	Table I	4.4.6
Lead content	-	-	Table I	4.4.9
Contrast ratio	-	D 2805	Table I	4.4.10
Chalk acceptance and removal	-	-	3.3.8	4.4.8
Appearance	-	-	3.3.10	4.4.3

3.3.2 Flexibility. The paint shall show no evidence of cracking when tested as specified in 4.4.2.

3.3.3 Skinning. The paint shall not skin after being stored for 2 months in a full container when tested as specified in table II.

3.3.4 Odor. The odor of the paint in the container and during and after application shall be characteristic of the solvents used when tested according to table II.

3.3.5 Brushing, flowing, spreading, and leveling properties. The paint, when tested according to 4.4.3, shall provide easy brushing, with good flowing, spreading, and leveling properties. Paints that have any noticeable pull or drag under a 5-inch wall brush or that show poor or only fair spreading and flowing properties are not acceptable.

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3.3.6 Color. The color of the paint shall be in accordance with Fed. Std. No. 595. The paint shall match color chip No. 37038 for the black and 34108 for the green when tested as specified in 4.4.4.

3.3.7 Fastness to light. The paint shall show no color variation or other change in appearance when tested as specified in 4.4.5.

3.3.8 Chalk acceptance and removal. The blackboard on which the paint is applied shall be easily marked by white chalk after drying for 24 hours and the chalk removed with light strokes of a blackboard eraser. The paint must have good chalk acceptance when examined as specified in 4.4.8.

3.3.9 Abrasion and moisture resistance. The paint film shall not show wear after 300 cycles of scrubbing, when tested as specified in 4.4.7.

3.3.10 Appearance of painted surface. The paint shall dry to a uniform, smooth, flat appearance under normal conditions of illumination and viewing. When the paint is tested as specified in 4.4.3 there shall be no shines, flashes, laps, or conspicuous brush marks.

3.4 Quantities. The paint shall be furnished in 1-quart or 1-gallon quantities, as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery and selected at random. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged or wrong type.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

4.3 Testing of the end item.

4.3.1 Lot. The paint shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the words "essentially the same conditions" shall be interpreted to mean a manufacturer's batch and defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.3.2 Sampling of the end item. For the purposes of sampling, the lot shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105.

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4.4 Test procedures.

4.4.1 The paint shall be tested according to the test methods indicated in table II and as specified hereinafter. Unless otherwise specified, standard testing conditions are a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in expressing the final result. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

4.4.2 Flexibility. Determine flexibility in accordance with Fed. Test Method Std. No. 141, method 6221, and as specified herein. Hold the tin panel by suitable means and apply paint to a dry film thickness of 76 um (0.003 in). Air dry the panel for 24 hours. Bend the panel over a 6.25 mm (1/4 in) mandrel, and inspect for compliance with 3.3.2.

4.4.3 Brushing, flowing, and leveling properties; appearance of dried film. Prepare a test panel of laminated composition wallboard measuring not less than 8 square feet in area. Apply two uniform brushed coats of plaster and wallboard primer-sealer to the smooth side of the panel. Allow the first coat to dry 24 hours before the second coat is applied. Also, allow the second coat to dry 24 hours. Place the panel in a vertical position and apply the well mixed paint with a 125 mm (5-in) wall brush as specified in Fed. Test Method Std. No. 141, method 2141, at a spreading rate of about 600 sq. ft. per gallon. In applying the paint, first brush horizontally across a section of the panel with back-and-forth strokes, then apply the tip of the brush to level the coat with light vertical strokes. Proceed to the next section in the same manner, always working to a wet edge. Determine whether or not the paint brushes easily and if it has good flowing and spreading properties. Let the panel dry for 24 hours. Observe whether the film has flowed out to a level, uniform appearance. Inspect for freedom from shiners, laps, and conspicuous brush marks and determine compliance with 3.3.5.

4.4.4 Color. A film of the paint shall be applied to a smooth, clean, plate glass panel to a dry film thickness of 76 um (0.003 in), and allowed to dry for 24 hours. Determine the color in accordance with ASTM method D 1729, and examine for compliance with 3.3.6.

4.4.5 Fastness to light. Apply a sufficient number of coats of paint to two similar ground panels to hide the surface completely. Allow each coat to dry for 24 hours. Expose one panel for 24 hours in a Fade-Ometer in accordance with ASTM method D 822. At the end of the exposure period, compare the exposed panel with the unexposed panel and determine compliance with 3.3.7.

4.4.6 Specular gloss. Prepare the panel by drawing down a 76 um (0.003 in) thick dry film of the paint on clear, polished, plane plate glass. Dry the panel in a horizontal position for 48 hours at a temperature of 23 ± 1°C and a relative humidity of 50 ± 4 percent. Measure the gloss as specified in ASTM method D 523, for compliance with table I.

4.4.7 Abrasion and moisture resistance. The paint shall be applied to scrub test panels to a dry film thickness of 88 um (0.0035 in). After 7 days, the panels shall be tested according to ASTM method D 2486 and evaluated for compliance with 3.3.9.

4.4.8 Chalk acceptance and removal. The paint panel prepared in 4.4.3 shall be used in this test. After the paint has dried for 24 hours, chalk lines shall be drawn across the full width of the painted area, using white chalk. Fifteen minutes later, these lines shall be removed using an eraser, and compliance with 3.3.8 evaluated.

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4.4.9 Lead content.

4.4.9.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be at least 10 inches long on the sealed portion of the penetration chart. The drawdown shall be cut into discs of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

4.4.9.2 Procedure. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength dispersive fluorescence spectrometer shall be as follows: (1)

Element	Analytical Line	Angle	Crystal	Detection	Collimeter	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60Kv 45Ma

Pulse height selection shall be used in all measurements, and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background, and the Molybdenum Compton scattered background from the X-ray tube.

4.4.9.3 Calculation.

$$R = \frac{I_{Pb} - \frac{I_{Pb}\ (\text{Background I}) + I_{Pb}\ (\text{Background II})}{2}}{I_{Mo}}$$

where I equals gross intensity. These results shall be compared to those obtained using a 0.06 percent lead standard made up from the same type of paint sample, and evaluated for compliance with table I.

4.4.10 Contrast ratio. Measure the contrast ratio at a dry film thickness of 76 um (0.003 in) as specified in ASTM method D 2805, and evaluate for compliance with table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2). Containers shall meet or exceed the requirements of DOT specifications in 49 CFR 171-189.

5.1.1 Level A. The paint shall be furnished in 1-quart metal cans or 1-gallon metal cans, as specified (see 6.2).

5.1.2 Commercial. The paint shall be packaged in cans in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2). Packing shall meet or exceed the requirements of DOT specifications in accordance with 49 CFR 171-189.

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

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5.2.1 Level A. Twelve 1-quart cans or four 1-gallon cans of paint, packaged as specified in 5.1, shall be packed in fiberboard boxes made from weather-resistant fiberboard with a bursting test strength of not less than 275 lb per square inch. The box flaps shall be secured with water-resistant adhesive applied to not less than 75 percent of the surface area of contact between the flaps, or with 3 inch wide waterproof tape applied to the full length of the seams and extending over the ends not less than 3 inches. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The one quart and one gallon paint, packaged as specified in 5.1, shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

6. NOTES

6.1 Intended use. The paint covered by this specification is intended primarily for surfacing blackboards. Although the paint is of the fast drying type, it is recommended that the paint be applied late in the day so as to have a fresh surface available for use in the morning.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 1.2.1).
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Size of container required (see 3.4).
(e) Marking required (see 5.4).

Civil Agency Coordinating Activities:

COMMERCE-NBS
FAA-SMS
GSA-FSS
NASA-JFK

Preparing activity:

GSA-FSS

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